1 Monte Carlo comparisons

1.1 Statistical comparison across independent realizations

For each benchmark and parametrization, the optimizers were compared using the same 30 independent input/noise realizations. The outcome for rule a in realization r was the final relative parameter error

$$E_{a,r} = 100 \frac{\|\hat{\boldsymbol{\theta}}_{a,r} - \boldsymbol{\theta}^\star\|_2}{\|\boldsymbol{\theta}^\star\|_2}. \tag{1}$$

For every unordered pair (a, b), we counted the realizations in which each rule produced the lower error and performed a two-sided exact paired sign test [1]. Under the null hypothesis of equal win probabilities, $W_{ab} \mid (W_{ab} + L_{ab}) \sim \text{Binomial}(W_{ab} + L_{ab}, 1/2)$; ties were excluded from the binomial count. Numerical ties were defined by $|E_{a,r} - E_{b,r}| \leq 10\ \text{eps}(\max\{1, E_{a,r}, E_{b,r}\})$, where $\text{eps}(x)$ denotes floating-point spacing at x. All finite outcomes, including extreme values, were retained. Each group contained eight applicable rules and 28 comparisons; Holm's correction [2] controlled the family-wise error rate at 0.05 separately within each benchmark–parametrization combination. Superiority denotes a significantly greater frequency of lower errors; the test does not establish a lower arithmetic mean or a practically important reduction. The legacy Monte Carlo files preserve pairing by realization column, but do not contain per-run hashes for retrospective verification. All applicable rules have 30 finite outcomes in these files.

Tables 1–4 report all pairwise outcomes. For Example 1/FSM, none of the 28 comparisons remained significant. For Example 1/PL, GHAM-2-I and LHAM-1-TIK outperformed WS-GGI in 28/30 and 25/30 realizations, respectively ($p_{\text{H}} = 2.43 \times 10^{-5}$ and 8.77×10^{-3}); the other 26 comparisons were inconclusive. For CSTR/FSM, 26 comparisons were significant. LHAM-1-TIK and LHAM-3-TIK each outperformed GI and all evaluated GHAM rules, but neither showed a significant advantage over Newton ($p_{\text{H}} = 0.585$ and 0.401). LHAM-1-TIK outperformed LHAM-3-TIK in 23/30 realizations ($p_{\text{H}} = 0.0157$), although their median paired error difference was only 1.97×10^{-4} percentage points in favor of LHAM-1-TIK. For CSTR/PL, 18 comparisons were significant; WS-GNI, RLS, GHAM-1-dH, and GHAM-2-dH each outperformed LHAM-1-TIK in 27/30 realizations ($p_{\text{H}} = 1.26 \times 10^{-4}$). Thus, the evidence supports specific update-rule advantages and does not establish one optimizer as uniformly superior. Inconclusive comparisons do not establish equivalence.

Table 1: Pairwise Monte Carlo results: Example 1, FSM. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: Newton [30/30]	–	17:13:0 1	13:17:0 1	15:15:0 1	15:15:0 1	15:15:0 1	17:13:0 1	17:13:0 1
2: GI [30/30]		–	13:17:0 1	15:15:0 1	15:15:0 1	15:15:0 1	15:15:0 1	14:16:0 1
3: GHAM-1-dH [30/30]			–	15:15:0 1	15:15:0 1	15:15:0 1	17:13:0 1	17:13:0 1
4: GHAM-2-dH [30/30]				–	15:15:0 1	15:15:0 1	15:15:0 1	15:15:0 1
5: GHAM-2-H [30/30]					–	15:15:0 1	15:15:0 1	15:15:0 1
6: GHAM-2-I [30/30]						–	15:15:0 1	15:15:0 1
7: LHAM-1-TIK [30/30]							–	16:14:0 1
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: RLS.

Table 2: Pairwise Monte Carlo results: Example 1, PL. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: RLS [30/30]	–	16:14:0 1	22:8:0 0.419	12:18:0 1	15:15:0 1	9:21:0 0.941	11:19:0 1	12:18:0 1
2: WS-GNI [30/30]		–	22:8:0 0.419	13:17:0 1	13:17:0 1	9:21:0 0.941	11:19:0 1	12:18:0 1
3: WS-GGI [30/30]			–	8:22:0 0.419	8:22:0 0.419	2:28:0 ↓ 2.4×10^{-5}	5:25:0 ↓ 0.00877	10:20:0 1
4: GHAM-1-dH [30/30]				–	16:14:0 1	9:21:0 0.941	11:19:0 1	12:18:0 1
5: GHAM-2-dH [30/30]					–	9:21:0 0.941	11:19:0 1	12:18:0 1
6: GHAM-2-I [30/30]						–	16:14:0 1	19:11:0 1
7: LHAM-1-TIK [30/30]							–	19:11:0 1
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: GHAM-2-H.

Table 3: Pairwise Monte Carlo results: CSTR, FSM. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: Newton [30/30]	–	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	13:17:0 0.585	19:11:0 0.401
2: GI [30/30]		–	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}
3: GHAM-1-dH [30/30]			–	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	29:1:0 ↑ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}
4: GHAM-2-dH [30/30]				–	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	1:29:0 ↓ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}
5: GHAM-2-H [30/30]					–	29:1:0 ↑ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}
6: GHAM-2-I [30/30]						–	1:29:0 ↓ 1.5×10^{-6}	1:29:0 ↓ 1.5×10^{-6}
7: LHAM-1-TIK [30/30]							–	23:7:0 ↑ 0.0157
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: RLS.

Table 4: Pairwise Monte Carlo results: CSTR, PL. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: RLS [30/30]	–	18:12:0 1	30:0:0 ↑ 5.2×10^{-8}	15:15:0 1	16:14:0 1	30:0:0 ↑ 5.2×10^{-8}	27:3:0 ↑ 1.3×10^{-4}	22:8:0 0.161
2: WS-GNI [30/30]		–	30:0:0 ↑ 5.2×10^{-8}	14:16:0 1	16:14:0 1	30:0:0 ↑ 5.2×10^{-8}	27:3:0 ↑ 1.3×10^{-4}	22:8:0 0.161
3: WS-GGI [30/30]			–	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
4: GHAM-1-dH [30/30]				–	16:14:0 1	30:0:0 ↑ 5.2×10^{-8}	27:3:0 ↑ 1.3×10^{-4}	22:8:0 0.161
5: GHAM-2-dH [30/30]					–	30:0:0 ↑ 5.2×10^{-8}	27:3:0 ↑ 1.3×10^{-4}	22:8:0 0.161
6: GHAM-2-I [30/30]						–	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
7: LHAM-1-TIK [30/30]							–	3:27:0 ↓ 1.3×10^{-4}
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: GHAM-2-H.

2 Ablation

2.1 Ablation: sensitivity to initialization

Initialization sensitivity was assessed separately using 30 randomly drawn starting points per benchmark and parametrization, with a fixed input/noise record. All rules received the same starting point within each paired trial; the saved metadata were checked for shared data and initialization. The final relative parameter error and two-sided exact paired sign test were defined as in Section 1.1, with a separate Holm family of 28 comparisons for each benchmark–parametrization combination and $\alpha = 0.05$. All finite extreme outcomes were included in the tests, including observations omitted from the boxplot display. Accuracy comparisons were conditional on both rules returning valid finite estimates; failed runs were not counted as accuracy wins for their competitors. The valid counts in the tables therefore qualify every superiority claim, and these tests alone do not establish a higher probability of successful completion or robustness across different data records.

Tables 5–8 contain all comparisons. The numbers of significant pairs were 18/28 and 14/28 for Example 1/FSM and Example 1/PL, and 22/28 and 24/28 for CSTR/FSM and CSTR/PL, respectively. For Example 1/PL, LHAM-3-TIK produced 30 finite outcomes and significantly lower errors than RLS, WS-GNI, GHAM-1-dH, GHAM-2-dH, WS-GGI, and GHAM-2-I on their jointly valid trials; its comparison with LHAM-1-TIK was inconclusive. Conversely, for CSTR/PL, RLS, WS-GNI, GHAM-1-dH, and GHAM-2-dH each significantly outperformed LHAM-3-TIK on jointly valid trials. RLS and WS-GNI returned only 14 and 16 finite estimates in each PL benchmark, whereas both LHAM variants returned 30. For Example 1/FSM, GHAM-2-H and GHAM-2-I frequently attained lower errors than the LHAM rules, but GHAM-2-I returned only 18 finite estimates and GHAM-2-H included a very large finite error. Frequent accuracy wins therefore do not imply protection against rare failures or extreme errors. For CSTR/FSM, both LHAM rules significantly outperformed all evaluated GHAM rules, while neither significantly outperformed Newton or GI. The observed initialization sensitivity is consequently specific to the benchmark, parametrization, and chosen robustness criterion.

Table 5: Pairwise initialization ablation results: Example 1, FSM. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: Newton [30/30]	–	22:2:0 ↑ 9.0×10^{-4}	4:26:0 ↓ 0.00143	9:19:0 0.61	5:25:0 ↓ 0.00455	1:17:0 ↓ 0.0029	28:2:0 ↑ 2.3×10^{-5}	29:1:0 ↑ 1.6×10^{-6}
2: GI [24/30]		–	4:20:0 ↓ 0.017	5:18:0 0.0956	3:21:0 ↓ 0.00439	1:16:0 ↓ 0.00439	11:13:0 1	12:12:0 1
3: GHAM-1-dH [30/30]			–	8:20:0 0.286	4:26:0 ↓ 0.00143	1:17:0 ↓ 0.0029	26:4:0 ↑ 0.00143	26:4:0 ↑ 0.00143
4: GHAM-2-dH [28/30]				–	16:12:0 1	12:6:0 0.952	19:9:0 0.61	19:9:0 0.61
5: GHAM-2-H [30/30]					–	15:3:0 0.0754	25:5:0 ↑ 0.00455	25:5:0 ↑ 0.00455
6: GHAM-2-I [18/30]						–	17:1:0 ↑ 0.0029	17:1:0 ↑ 0.0029
7: LHAM-1-TIK [30/30]							–	29:1:0 ↑ 1.6×10^{-6}
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: RLS.

Table 6: Pairwise initialization ablation results: Example 1, PL. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: RLS [14/30]	–	9:5:0 1	13:1:0 ↑ 0.0293	11:2:0 0.247	9:3:0 1	11:2:0 0.247	9:5:0 1	1:13:0 ↓ 0.0293
2: WS-GNI [16/30]		–	15:1:0 ↑ 0.00934	13:2:0 0.103	10:4:0 1	13:2:0 0.103	10:6:0 1	1:15:0 ↓ 0.00934
3: WS-GGI [30/30]			–	2:27:0 ↓ 3.7×10^{-5}	1:25:0 ↓ 2.0×10^{-5}	15:13:0 1	1:29:0 ↓ 1.6×10^{-6}	2:28:0 ↓ 2.1×10^{-5}
4: GHAM-1-dH [29/30]				–	6:20:0 0.112	25:2:0 ↑ 1.1×10^{-4}	17:12:0 1	2:27:0 ↓ 3.7×10^{-5}
5: GHAM-2-dH [26/30]					–	23:1:0 ↑ 6.3×10^{-5}	15:11:0 1	2:24:0 ↓ 2.0×10^{-4}
6: GHAM-2-I [28/30]						–	1:27:0 ↓ 5.8×10^{-6}	1:27:0 ↓ 5.8×10^{-6}
7: LHAM-1-TIK [30/30]							–	10:20:0 0.889
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: GHAM-2-H.

Table 7: Pairwise initialization ablation results: CSTR, FSM. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: Newton [30/30]	–	15:15:0 1	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	15:15:0 1	17:13:0 1
2: GI [30/30]		–	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	15:15:0 1	16:14:0 1
3: GHAM-1-dH [30/30]			–	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
4: GHAM-2-dH [30/30]				–	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
5: GHAM-2-H [30/30]					–	18:12:0 1	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
6: GHAM-2-I [30/30]						–	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
7: LHAM-1-TIK [30/30]							–	30:0:0 ↑ 5.2×10^{-8}
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: RLS.

Table 8: Pairwise initialization ablation results: CSTR, PL. Each upper-triangular entry gives row wins:losses:ties, followed by the Holm-adjusted p-value. ↑ denotes a significant advantage for the row rule; ↓ for the column rule. An unmarked p-value is inconclusive at $\alpha = 0.05$.

Rule [valid/30]	1	2	3	4	5	6	7	8
1: RLS [14/30]	–	4:10:0 0.359	14:0:0 ↑ 0.00122	13:1:0 ↑ 0.00916	11:2:0 0.0851	12:0:0 ↑ 0.00293	14:0:0 ↑ 0.00122	14:0:0 ↑ 0.00122
2: WS-GNI [16/30]		–	16:0:0 ↑ 4.0×10^{-4}	13:3:0 0.0851	9:6:0 0.607	14:0:0 ↑ 0.00122	16:0:0 ↑ 4.0×10^{-4}	16:0:0 ↑ 4.0×10^{-4}
3: WS-GGI [30/30]			–	0:30:0 ↓ 5.2×10^{-8}	0:28:0 ↓ 1.6×10^{-7}	28:0:0 ↑ 1.6×10^{-7}	0:30:0 ↓ 5.2×10^{-8}	0:30:0 ↓ 5.2×10^{-8}
4: GHAM-1-dH [30/30]				–	1:27:0 ↓ 3.0×10^{-6}	28:0:0 ↑ 1.6×10^{-7}	30:0:0 ↑ 5.2×10^{-8}	30:0:0 ↑ 5.2×10^{-8}
5: GHAM-2-dH [28/30]					–	26:0:0 ↑ 4.5×10^{-7}	28:0:0 ↑ 1.6×10^{-7}	28:0:0 ↑ 1.6×10^{-7}
6: GHAM-2-I [28/30]						–	0:28:0 ↓ 1.6×10^{-7}	0:28:0 ↓ 1.6×10^{-7}
7: LHAM-1-TIK [30/30]							–	0:30:0 ↓ 5.2×10^{-8}
8: LHAM-3-TIK [30/30]								–

The valid count includes finite usable estimates; a paired test uses only jointly valid runs. Its effective binomial sample size is wins plus losses. All finite extreme values are retained. Blank lower-triangular cells omit symmetric duplicates. Unavailable rule: GHAM-2-H.

References

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